Investigating the usage of Unsupervised Machine Learning in classifying Astral Objects

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Introduction

In astronomy we seek to better understand our universe through observing various celestial objects that let us gain insight into how our universe formed. Through analysing measurements of these objects which include wavelengths of light they reflect, their redshift and luminosity, we seek to categorize them as a galaxy, star or quasar. Various supervised and unsupervised methods have been used to tackle this problem, unsupervised methods have been used in literature to tackle these problems (Gao 2023; Cody et al. 2024). A common limitation both methods have is the quality of the data collected since it is extracted from images taken by telescopes (Zeraatgari et al. 2024).

Dataset

Our dataset is a table of 10000 celestial objects that comprises of 11 data points. Object ID and the type of body the object is (galaxy, star or quasar). Right Ascension and Declination, the longitude and latitude equivalent in celestial measurements. 5 wavelength bands labeled u ($\lambda = 335$ nm, Ultraviolet), g ($\lambda = 469$ nm, Green/Blue), r ($\lambda = 617$ nm, Red), i ($\lambda = 748$ nm, Near-Infrared) and z ($\lambda = 893$ nm, Infrared), which are logarithmic readings of an object's brightness in that band where a lower number means a more intense reading. And lastly Redshift which is a non-negative number describing how much the wavelength of light was stretched when measuring, it is due to the celestial objects traveling away relative to Earth causing the wavelengths to be increased, a higher redshift corresponds to an object further away and traveling at a higher velocity away from Earth.

Methodology

In this section we present our feature selection, data visualisation with dimensionality reduction and the various models we consider.

Feature Selection

Within the dataset there are NaNs, data entries that are set to -9999, which are empty data points that provide no meaning to our models. We replace such data points with the median of that feature before proceeding with choosing which features to remove and how to transform the remaining features before training our models.

Firstly, we remove the type of object from the dataset as this classification problem is being done with unsupervised learning. Next we remove Object ID since it is the index of the object observed and has no correlation with the type of object nor the other observed data. Lastly we remove Right Ascension and Declination since the location of the celestial object in the sky does not correlate to what type of object it is (Cody et al. 2024).

We then take the differences between adjacent bands and get the features u-g, g-r, r-i and i-z. Taking these differences reduces overlap between stars and other celestial objects and provides the models with more information during training (Zeraatgari et al. 2024). Since we use the K-means machine learning method later, we also normalize the differences and redshift data to ensure the numerical data is roughly linear, this is important especially for redshift since galaxies and quasars have considerably bigger redshift than stars (Cody et al. 2024).

Data visualisation

We first use PCA (Principal Component Analysis) and UMAP to do dimensionality reduction on the dataset, the plots are presented in Figure 1. We observe 3 clusters and initialise our models (except HDBSCAN) accordingly.

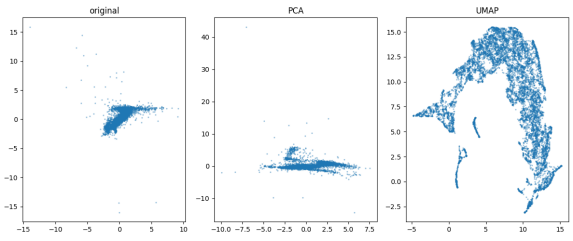


Figure 1: Visualising raw data, data after PCA and data after UMAP

K-means++

K-means++ is an unsupervised algorithm that clusters data by shifting centroids around such that the sum of the dis-

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tances between data points and their nearest centroid is minimised. The initial centroid positions are spread out sufficiently using randomness.

GMM (Gaussian Mixture Models)

GMM is an unsupervised learning algorithm that assumes the data is distributed by a multivariate gaussian distribution. For each data point, we compute the probability it belongs to each gaussian and then shift the mean and variance accordingly, this algorithm terminates if the parameters converge or when the logarithm of the probability that the data is described by the gaussians is 1 (Fieguth 2022, 440-441).

Agglomerative Clustering

Aggromatice clustering is an unsupervised algorithm which deals with combining clusters with maximum similarity or minimum similarity at each step until the desired number of clusters is produced. Initialise each point in the dataset as its own cluster, merge 2 clusters together based on a predetermined metric. Repeat the above step on all remaining clusters until the distance between clusters is large. An example of a metric is identifying the 2 clusters which have the smallest distance across all pairs of points between them (Miyamoto 2022, 20-21).

HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise)

HDBSCAN is a modification of DBSCAN, an unsupervised algorithm. Initialise DBSCAN by selecting a fixed number of clusters and a random datapoint. First, it adds all other data points within a fixed distance to the current cluster, repeat until no other points can be added. Repeat this on all points not in any cluster, classify points that are not near any cluster as noise. The algorithm terminates once all points are assigned to a cluster or designated as noise (Ester et al. 1996). In DBSCAN the distance is predetermined making it costly to determine the optimal value, HDBSCAN improves on this by varying over values to find an optimal distance, we initialise it with 50 clusters (McInnes and Healy 2017).

Results

To evaluate the effectiveness of each model, we use the conductance score by turning the clusters that the models produce into a graph. We are able to transform the continuous parameters into a graph and use conductance scores to evaluate them (Shi and Malik 2000; Arias-Castro, Pelletier, and Pudlo 2012).

	K-means++	GMM	Agglomerative	HDBSCAN
Conductance score	3.2×10^{-3}	4.1×10^{-3}	0.0	0.0

Table 1: Conductance score of models

When we initialise the models (except HDBSCAN) with 2 to 7 clusters, the conductance score is the lowest at 3 clusters, this agrees that we expect 3 clusters from Figure 1.

From the conductance scores, HDBSCAN and agglomerative clustering perform the best at clustering the data. However, when comparing their implementation, HDBSCAN is much more costly than Agglomerative clustering due the need of deciding the number of clusters before hand. We therefore choose agglomerative clustering as our final model and compare it to the true visualisation of the original data.

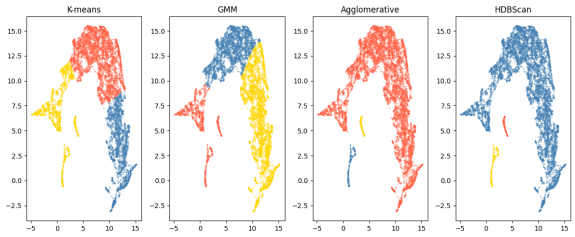


Figure 2: Plots of various clustering models

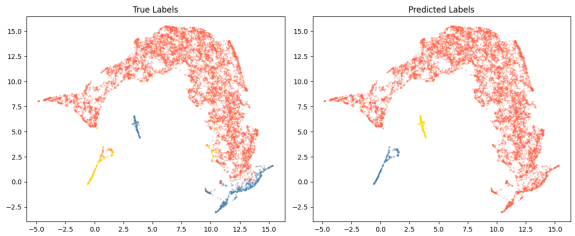


Figure 3: True clusters and predicted clusters

Lastly we validate our model using Adjusted Rand Index (ARI) as presented in Figure 3, achieving a score of 0.457. The model cleanly groups the data into 3 neat clusters, however there is some mixing of the clusters in the True Labels plot that the model is unable to differentiate in Figure 3. While the agglemorative clustering model preserves the overall structure of the clusters with good accuracy, it is unable to differentiate the mixing of the clusters due to the close proximity.

Comparing the True Labels plot against Figure 2, K-means++ and GMM fail to produce the desired clusters. This is likely due to the data not being spherical shaped nor being generated by gaussian distributions. This is attributed to the irregular shapes of the data causing both algorithms to fail, while agglomerative clustering and HDBSCAN are able to adapt it and produce meaningful clusters similar to the true data distribution.

Agglomerative clustering and HDBSCAN also fall short in being unable to distinguish the mixing between the true data. Since this problem is using unsupervised learning, we are unable to train our models meaningfully to handle it. If a human was given the data presented as in Figure 1, they will likely reach the same conclusion as these 2 models.

Conclusion and future work

We have produced an agglomerative model with good accuracy at clustering celestial objects. We observe unsupervised models which do not depend on the innate structure of the data perform better than those that do. Future work includes implementing supervised learning on this dataset and compare the performance of those models against our model.

AI Declaration

In brainstorming for which unsupervised models to consider, AI was used for the following aspects:

1. Brainstorming what unsupervised clustering algorithms to use.
2. Finding different methods to evaluate them.

In typing the report, AI was used for the following aspects:

1. Searching for publicly accessible papers to cite that match our research question.
2. Searching for relevant textbooks and papers that describe the algorithms that we consider.
3. Providing the appropriate LaTeXcode for figure insertion, table creation and citation methods.

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Contributions

Listed according to the order at the top of this report.

1. Tian Ruixi: Handled the Presentation slides.
2. Arya Bharathwaj Vijay: Handled the coding aspect of the model with Nigel.
3. Ewan Wijaya: Handled typing of the report.
4. Nigel Jarod Yong Tshin Kai: Handled the coding aspect of the model with Arya and exported the models.